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1. (a) Find all normal subgroups of the symmetric group S_3 .

$$* |S_3| = 6$$

$\Rightarrow$ Possible orders of subgroups of S_3 are 1, 2, 3, and 6

$$S_3 = \{(e), (123), (132), (12), (13), (23)\}$$

$\Rightarrow$ Subgroups: $\{e\}, S_3, A_3, \{e, (12)\}, \{e, (13)\}, \{e, (23)\}$

$$\boxed{\{e\}, \text{ since the identity commutes with everything}}$$

$$\boxed{S_3, \text{ since the conjugation of elements in } S_3 \text{ will, by closure, also be an element of } S_3}$$

$$\boxed{A_3, \text{ since it is true that } A_n \triangleleft S_n}$$

Remaining subgroups: $\{e, (12)\}, \{e, (13)\}, \{e, (23)\}$

$$\Rightarrow (13)(12)(13)^{-1} = (13)(12)(13) = (23) \notin \{e, (12)\}$$

$$\Rightarrow (12)(13)(12)^{-1} = (12)(13)(12) = (23) \notin \{e, (13)\}$$

$$\Rightarrow (13)(23)(13)^{-1} = (13)(23)(13) = (12) \notin \{e, (23)\}$$

$\Rightarrow$ All choices of normal subgroups for S_3 have been exhausted

- (b) In the group $U(100)$, let $H = \langle 21 \rangle = \{1, 21, 41, 61, 81\}$. Find $|9H|$ and $|13H|$ in the factor group $U(100)/H$.

$$*(aH)^n = a^n H \Rightarrow a^n \in H \text{ if } |a| = n$$

$$9^2 = 81 \pmod{100} = 81 \in H$$

$$\Rightarrow \boxed{|9H| = 2}$$

$$13^4 = 28,561 \pmod{100} = 61 \in H$$

$$\Rightarrow \boxed{|13H| = 4}$$

2. (a) Prove that a factor group of a cyclic group is cyclic.

Proof. Let $G = \langle a \rangle$ where $a \in G$ be a group and $H \triangleleft G$. We need to show that $G/H = \langle bH \rangle$ for some $bH \in G/H$. Since G is cyclic and $b \in G$,

$$\begin{aligned} bH &= (a^i)H \text{ for some } i \in \mathbb{Z} \\ &= (aH)^i. \end{aligned}$$

This shows that every element in G/H is a power of the left coset aH . Hence, $G/H = \langle aH \rangle$ given that $a \in G$ where G is cyclic. $\square$

- (b) Let k be a positive divisor of a positive integer n . To what familiar group is $\mathbb{Z}_n/\langle k \rangle$ isomorphic?

$$\begin{aligned} \text{ex) } n &= 10, k = 2 \\ \Rightarrow |\mathbb{Z}_{10}| &= 10, |\langle 2 \rangle| = 5 = \frac{10}{2} \end{aligned}$$

Consider $|\mathbb{Z}_n/\langle k \rangle|$. By Lagrange's Theorem, this is the same as $\frac{|\mathbb{Z}_n|}{|\langle k \rangle|} = \frac{n}{(n/k)} = k$.

We know $\mathbb{Z}_n$ for any $n \geq 1$ is cyclic and, by (a), $\mathbb{Z}_n/\langle k \rangle$ is cyclic. Given that $k > 0$, $\mathbb{Z}_k$ is also cyclic and $|\mathbb{Z}_k| = k$. So, it must be that $\mathbb{Z}_k \cong \mathbb{Z}_n/\langle k \rangle$

3. (a) Suppose H is a subgroup and N is a normal subgroup of a group G . Prove that $HN := \{hn \mid h \in H, n \in N\}$ is a subgroup of G .

Proof. Let G be a group, $H \leq G$, and $N \triangleleft G$. Let $HN := \{hn \mid h \in H, n \in N\}$. By the One-Step Subgroup Test, we want to show that $(h_1n_1)(h_2n_2)^{-1} \in HN$ where, by closure, $h_1h_2 \in H$ and $n_1n_2 \in N$. By the Socks-Shoes Property,

$$\begin{aligned} (h_1n_1)(h_2n_2)^{-1} &= (h_1n_1)(n_2^{-1}h_2^{-1}) \\ &= h_1(n_1n_2^{-1})h_2^{-1} \\ &= h_1n_3h_2^{-1} \text{ where } n_3 = n_1n_2^{-1} \in N. \end{aligned}$$

Since $H \leq G$, $H \subseteq G$ and $h_2^{-1} \in G$. Since N is normal, we know that $gN = Ng \forall g \in G$. So, $\exists$ some $n_4 \in N$ s.t. $n_3h_2^{-1} = h_2^{-1}n_4$.

$$\begin{aligned} (h_1n_1)(h_2n_2)^{-1} &= h_1(n_3h_2^{-1}) \\ &= h_1(h_2^{-1}n_4) \text{ for some } n_4 \in N \\ &= (h_1h_2^{-1})n_4 \\ &= h_3n_4 \text{ where } h_3 = h_1h_2^{-1} \in H. \end{aligned}$$

Thus, by the One-Step Subgroup Test, $HN \leq G$. □

- (b) Let N be a normal subgroup of G , and let H be a subgroup of G that contains N . Prove that H is normal in G if and only if H/N is normal in G/N .

Proof. Let G be a group, $N \triangleleft G$, and $H \leq G$ such that $N \subseteq H$.

($\Rightarrow$) Suppose H is normal in G . That means $gH = Hg$ and $gHg^{-1} \subseteq H \forall g \in G$. We want to show that H/N is normal in G/N , that is $(gN)H/N(gN)^{-1} \subseteq H/N$. Let $hN \in H/N$ and consider the expression

$$gNhN(gN)^{-1} = (ghg^{-1})N.$$

Since H is normal, $\exists$ some $h' \in H$ s.t. $ghg^{-1} = h'$. So, $gNhN(gN)^{-1} = h'N \in H/N$ which shows that $gNH/N(gN)^{-1} \subseteq H/N$. Hence, by Theorem 9.1, $H/N \triangleleft G/N$.

($\Leftarrow$) Suppose $H/N \triangleleft G/N$. Using the implication above, this means $(ghg^{-1})N \in H/N \forall g \in G, h \in H$. Thus, $ghg^{-1}n = h_0n$ for some $h_0 \in H$. But, since $N \subseteq H$, $n \in H$. So,

$$\begin{aligned} ghg^{-1}n(n^{-1}) &= h_0(n^{-1}) \\ \Rightarrow ghg^{-1}(nn^{-1}) &= h_0n^{-1} \\ \Rightarrow ghg^{-1} &= h' \text{ where } h' = h_0n^{-1} \in H. \end{aligned}$$

So, $ghg^{-1} \in H$ which shows $gHg^{-1} \subseteq H$ and that $H \triangleleft G$. □

4. (a) If N is a normal subgroup of G and $|G/N| = m$, show that $x^m \in N$ for all elements $x \in G$.

Proof. Let G be a group, $N \triangleleft G$, and $|G/N| = m$. Take some $xN \in G/N$ where $x \in G$. We know that $(xN)^m = (x^m)N$. However, since $|G/N|$ is finite, by Corollary 4 of Lagrange's Theorem,

$$\begin{aligned}(xN)^m &= (x^m)N \\ &= eN \\ &= N.\end{aligned}$$

So, $(x^m)N = N$ which, by Lemma 2 from Chapter 7, shows that $x^m \in N \forall x^m \in G$. $\square$

- (b) If N is a normal subgroup of a group G and $|N| = 2$, show that N is contained in the center of G .

Proof. Let G be a group, $N \triangleleft G$, and $|N| = 2$ such that $N = \{e, n\}$. Since N is normal, $gNg^{-1} \subseteq N \forall g \in G$ which makes $gng^{-1} \in N$. If $gng^{-1} = e$,

$$\begin{aligned}(g^{-1})gng^{-1}(g) &= (g^{-1})eg \\ \Rightarrow n &= e.\end{aligned}$$

But this implies $|N| = 1$ which contradicts our assumption. So, gng^{-1} must be n . Thus,

$$\begin{aligned}gng^{-1}(g) &= n(g) \\ \Rightarrow gn &= ng.\end{aligned}$$

This is precisely the definition of commutativity between elements of G and N . We know the identity is always in the center of a group. So, all elements of N commute with G and $N \in Z(G)$. $\square$

5. Suppose G is a finite group, and H is a normal subgroup of G . If the factor group G/H contains an element aH of order n , show that G must also contain an element of order n .

Proof. Let G be a finite group and $H \triangleleft G$. Take $aH \in H/G$ s.t $|aH| = n$. This means $(aH)^n = a^nH = H$. So, by Lemma 2 from Chapter 7, $a^n \in H$. But, since H is a subgroup of G , $H \subseteq G$ which in turn makes $a^n \in G$. $\square$

6. If a group G is non-Abelian, show that $\text{Aut}(G)$ is not cyclic.

Proof. Let G be a non-Abelian group and, by contradiction, suppose $\text{Aut}(G)$ is cyclic. Since $\text{Inn}(G)$ only contain automorphisms, it must be a subgroup of $\text{Aut}(G)$. By the Fundamental Theorem of Cyclic Groups, this means $\text{Inn}(G)$ is also cyclic. By Theorem 9.4, $\text{Inn}(G) \cong G/Z(G)$ where $Z(G)$ is the center of G . This implies that $G/Z(G)$ is also cyclic. But, by Theorem 9.3, $G/Z(G)$ being cyclic is true when G is Abelian which contradicts our assumption. So, $\text{Aut}(G)$ is not cyclic. $\square$